

Classifying Endurance Runners by Kinematics Strengthens Relationships of Muscle-Tendon Unit Function with Running Economy and Performance

Yuuri Eihara¹, Momoka Kinoshita¹, Koshi Jotoku¹, Sumiaki Maeo^{1,2}, Hiroaki Kanehisa^{2,3}, and Tadao Isaka^{1,2}

¹Faculty of Sport and Health Science, Ritsumeikan University, Shiga, JAPAN; ²Institute of Advanced Research for Sport and Health Science, Ritsumeikan University, Shiga, JAPAN;

³National Institute of Fitness and Sports in Kanoya, Kagoshima, JAPAN

Running Title: KINEMATIC CLUSTERING OF ENDURANCE RUNNERS

Address for correspondence: Sumiaki Maeo, PhD, Faculty of Sport and Health Science, Ritsumeikan University, 1-1-1 Nojihigashi, Kusatsu, Shiga, 525-8577, Japan; Tel: +81(0) 077-599-4135; E-mail: s-maeo@fc.ritsumei.ac.jp

Conflict of Interest and Funding Source: The experiment was supported by JST SPRING, Grant Number JPMJSP2101. The authors declare no conflicts of interest.

Acknowledgements: The experiment was supported by JST SPRING, Grant Number JPMJSP2101. The authors declare no conflicts of interest. The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The results of the present study do not constitute endorsement by the American College of Sports Medicine. YE conceived and designed the study. All authors contributed to data collection, analysis, and/or interpretation of the results. YE, SM, and HK drafted the manuscript. All authors read and approved the final version of the manuscript. This study was approved by the Ethics

Committee of Ritsumeikan University (BKC-LSMH-2023-087). Written informed consent was obtained from each participant. The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Copyright © 2026 The Author(s). Published by Wolters Kluwer Health, Inc. on behalf of the American College of Sports Medicine. This is an open-access article distributed under the terms of the Creative Commons Attribution-Non Commercial-No Derivatives License 4.0 (CCBY-NC-ND), where it is permissible to download and share the work provided it is properly cited. The work cannot be changed in any way or used commercially without permission from the journal

ABSTRACT

Purposes: Running kinematics and sex may alter the primary muscle and tendon demands for propulsion and shock absorption, potentially modifying how muscle-tendon unit (MTU) function relates to the energy cost of running and performance. This study examined whether clustering runners by running kinematics strengthens these relationships and how they differ by sex.

Methods: One hundred twenty endurance runners (60 females) underwent assessments of MTU function, including slow (30°/s) and fast (180°/s) isokinetic knee extension (KE) and flexion (KF) torque and performance in countermovement, drop, and rebound jumps. These variables were summarized using principal component analysis. Running kinematics and energy cost of running were assessed at 11 km/h and 80% of $\dot{V}O_{2max}$, and the season's best time (World Athletics [WA] score) was recorded. Runners were clustered via k-means based on foot strike angle and KE/KF range of motion. Stepwise regression analyses tested whether MTU-related associations differed between cluster-specific and whole-cohort models. **Results:** K-means identified three or four clusters per sex at each running velocity condition. In whole-cohort analyses for each sex, only weak to fair correlations were found between MTU function and either energy cost ($r = -0.204$ to 0.267) or WA score ($r = -0.276$ to 0.175). Clustering significantly ($P < 0.05$) strengthened these relationships in a cluster-specific manner (adjusted $R^2 = 0.159$ to 0.595). This pattern was more evident at 80% of $\dot{V}O_{2max}$, particularly in male clusters. Across conditions, MTU-related variables were selected in a greater proportion of female clusters (57.1%) than male clusters (31.2%). **Conclusions:** Kinematic-based clustering strengthens sex- and cluster-specific associations of MTU function with the energy cost of running and performance, supporting more individualized performance enhancement strategies.

Key Words: CLUSTER ANALYSIS, ENERGY COST OF RUNNING, LONG-DISTANCE
RUNNING, MUSCLE STRENGTH, JUMP PERFORMANCE

ACCEPTED

INTRODUCTION

Participation in running events has expanded markedly, from 1.8 million in 2001 to 8 million in 2018 (1). Success in endurance running is largely determined by aerobic parameters such as running economy and maximal oxygen uptake ($\dot{V}O_{2\max}$) (2). However, increasing $\dot{V}O_{2\max}$ in trained endurance runners can be challenging (3). In contrast, running economy, defined here as the metabolic or energy cost of running at a given speed (4), is influenced by modifiable factors including muscle-tendon unit (MTU) function (5, 6), running biomechanics (7-9), and physiological characteristics (10-12). Among these, MTU function, such as muscle strength and jump performance, can be improved relatively easily by incorporating strength training into running programs (13). Accordingly, many studies have investigated factors affecting running economy and performance, as well as strength training modalities that optimize these outcomes (14, 15).

Lower-limb muscles, in interaction with tendon behaviour, are essential for achieving superior running economy and performance. The quadriceps femoris attenuates impact forces at touchdown (16, 17), while the hamstrings contribute to propulsion during the late stance (18, 19). Co-contraction of these muscle groups increases knee and leg stiffness (20, 21), enabling the efficient use of impact forces through the stretch-shortening cycle (SSC) (5) and, consequently, improvement in running economy (22). Furthermore, interaction between the triceps surae and Achilles tendon during running mirrors that seen in jumping (23), and such interaction has been linked to economical running (12, 24). Indeed, resistance and plyometric training aimed at enhancing muscle strength and jump performance can improve running economy and performance (13, 25, 26).

Nevertheless, responses to resistance and plyometric training aimed at improving running economy vary considerably among individuals (27), suggesting that the importance of specific MTU function for running economy and performance differs between runners. This interpretation is supported by previous evidence indicating that relationships between movement patterns and performance outcomes may differ across subgroups (22). Sex is one factor likely to influence these relationships between MTU function and both running economy and performance. Female runners often (28, 29), but not always (30), exhibit lower knee flexor and higher knee extensor activation than males when running at the same absolute velocity (e.g., 12 km/h). This pattern is thought to reflect greater knee flexion (KF) angle and increased quadriceps activity at touchdown in females (31-33). Consequently, the association between knee extension (KE) torque and running economy or performance may be more pronounced in females than in males.

Even within the same sex, running kinematics may influence which MTU functions are most relevant to running economy and performance. Rearfoot strikers show greater biceps femoris activation than forefoot strikers at the same absolute velocity (34-36), and runners with a larger knee extension/flexion range of motion (KE/KF ROM) activate the rectus femoris more during propulsion [27]. Conversely, forefoot strikers with smaller KE/KF ROM tend to rely more on SSC utilization (i.e., effective elastic energy storage and return and stretch-related potentiation) of the lower leg MTU (20, 35). Therefore, grouping runners with similar kinematic profiles (e.g., foot strike angle and KE/KF ROM) may strengthen the observed relationships of MTU function with running economy and performance. For example, clusters characterized by a dorsiflexed foot strike (rearfoot) may show stronger associations between KF strength and running economy and/or performance.

The k-means clustering method groups participants based on multivariate profiles into distinct clusters and is particularly useful when the same set of variables is measured for all participants (e.g., running kinematics), enabling pattern-based (parallel) clustering across individuals (37, 38). In this study, we applied k-means clustering to classify endurance runners according to foot strike angle and KE/KF ROM at the same running velocity. Because racing speeds vary between individual runners, we examined relationships between MTU function and running economy and/or performance at both the same absolute speed (11 km/h) and the same relative speed (80% of the velocity of $\dot{V}O_{2\max}$ [$v\dot{V}O_{2\max}$]). The purposes of this study were 1) to examine relationships of MTU function with running economy and performance in the whole cohort and within each kinematic cluster, and 2) to determine how these relationships differ between sexes at absolute and relative velocities. We hypothesized that 1) clustering would strengthen these relationships compared with non-clustered analyses, and 2) clusters showing strong KE strength-economy and/or performance relationships would be more prevalent in females than in males.

METHODS

Participants

One hundred twenty trained (Tier 2–3 (39)) endurance runners (60 females) voluntarily participated in this study (Table 1). Participants were required to meet the following inclusion criteria: 1) age 18–50 years; 2) season's best time in a 5 km or longer race with World Athletics (WA) points of > 54 for males (e.g., full-marathon < 4:00:05) and > 259 for females (e.g., full-marathon < 4:30:02) (40); 3) body mass index < 24 kg/m²; 4) no musculoskeletal injuries requiring modified training within the past 3 months and no injuries within the past month; and 5) running more than twice per week during the preceding 3 months. Running performance was

expressed as the WA score converted from the season's best race time. Participants were stratified by age group (18–30, 31–40, and 41–50 years), with 40 participants per group and a 1:1 male-to-female ratio. This study was approved by the ethics committee of Ritsumeikan University (BKC-LSMH-2023-011) and conducted in accordance with the Declaration of Helsinki. All participants provided written informed consent after being informed of the study purpose, procedures, and potential risks.

Experimental design and procedure

Participants attended two laboratory sessions separated by 24–168 h, under controlled ambient temperature (21.0–22.7°C). They refrained from strenuous exercise and alcohol for 24 h, and caffeine for 6 h before each visit. The following variables were measured: Session 1) body composition, KE/KF strength, jump performance, and $\dot{V}O_{2\max}$; Session 2) body composition, running kinematics, and energy cost of running (ECR). Season's best performance within the past year was obtained from official records. Body composition data from Session 1 were used to scale $\dot{V}O_{2\max}$, and from Session 2 to normalize ECR.

Body composition. Height and body mass were measured using an automatic scale with a stadiometer (WB-510, TANITA Corp., Tokyo, Japan). Body fat percentage, muscle mass, and fat-free mass were measured using a multi-frequency, segmental bioelectrical impedance analyzer (InBody770, InBody Japan Co., Ltd., Tokyo, Japan), which has demonstrated good validity compared with dual-energy X-ray absorptiometry (41, 42).

KE/KF torque. Isokinetic KE/KF concentric torque of the right leg was assessed in a seated position (hip flexion 85°) using a dynamometer (Biodex System IV, Biodex Medical Systems, Shirley, NY). The seat was adjusted to align the lateral tibial condyle with the dynamometer axis of rotation, and the trunk and tested thigh were secured with straps. Gravity correction was

applied for the static position at approximately 45° of knee flexion. Participants performed a warm-up of three isometric KE/KF contractions at 90° knee flexion at 50%, 70%, and 90% of maximal effort, followed by practice isokinetic contractions in both slow (30°/s) and fast (180°/s) conditions. These practice trials consisted of 2–3 complete cycles of KE and KF, from 90° KF to 0° full extension and back to 90° flexion, performed at 50% effort. Maximal isokinetic concentric torque was then measured during two complete cycles at each angular velocity, with a rest period of 60 s between velocity conditions. The highest torque value for KE and KF at each velocity was used for analysis.

Jump performance. CMJ, 30 cm drop jump, and three consecutive rebound jump tests were performed three times each. Jump height and ground contact time were recorded using an optical measurement system (OptoJump Next, Microgate, Bolzano, Italy), which has been validated previously (43, 44). For the CMJ, participants performed a rapid squat followed by a maximal vertical jump with an arm swing. The highest jump height was used for analysis. For the drop jump, participants stood on a 30 cm box, stepped off, landed, and rebounded to maximal height with an arm swing while minimizing ground contact time. For the rebound jumps, participants performed three consecutive maximal vertical jumps with an arm swing and minimal contact time between jumps. For both the drop and rebound jumps, a jump index (jump height divided by contact time) was calculated, and the trial with the highest jump index was used for analysis. If two or three trials showed the same highest jump index, the trial with the greatest jump height was selected.

$\dot{V}O_{2max}$. $\dot{V}O_{2max}$ was measured on a treadmill with no incline (Valiant Ultra; Lode BV, Groningen, Netherlands) using a breath-by-breath gas analyzer (AE-310s, Minato Medical Science, Osaka, Japan) and masks (AMA104 and AMA512, Minato Medical Science, Osaka,

Japan). Heart rate was recorded using a heart rate sensor (Polar H10, Polar, Kempele, Finland). The gas analyzer was calibrated before each session with reference gases (16% O₂ and 5% CO₂). The repeatability of measurements with this breath-by-breath gas analysis system has been confirmed and is reported in Supplemental Table 1, <http://links.lww.com/MSS/D402> (Supplemental Digital Content). As a warm-up, participants ran on the treadmill at 8 and 11 km/h for 4 min each, separated by a 2-min rest, consisting of a 1-min walk at 3.2 km/h followed by 1 min of passive rest. After an additional 2-min rest, participants performed a continuous incremental test starting at 8 km/h, with velocity increased by 1 km/h every 1-min up to 14 km/h, and thereafter by 0.6 km/h until volitional exhaustion, defined as the inability to maintain the required running velocity. $\dot{V}O_{2max}$ was defined as the highest 30-s moving average. Attainment of $\dot{V}O_{2max}$ was confirmed if all of the following criteria were met: heart rate $\geq$ 90% of age-predicted maximum (220 - age); respiratory exchange ratio (RER) $\geq$ 1.10 (26); or an increase of $\dot{V}O_2 \leq 1$ mL/kg/min with increased workload (45). All participants completed the test until volitional exhaustion, and all predefined criteria were met. The velocity associated with $\dot{V}O_{2max}$ ($v\dot{V}O_{2max}$) was calculated by accounting for fractional completion of the final stage (46). For example, if a participant completed 30 s of the final stage at 13 km/h, $v\dot{V}O_{2max}$ was calculated as 13.5 km/h (13 [velocity of the final stage] + 1 [increment rate] $\times$ 30/60 [fractional time]).

Energy cost and allometric scaling. The ECR was measured on a 0% gradient treadmill (Aspire; Life Fitness, Illinois, USA) using the same gas analyzer and mask as for the $\dot{V}O_{2max}$ measurement, at 11 km/h and 80% of $v\dot{V}O_{2max}$. A 0% incline was selected because recent evidence indicates that, at submaximal running speeds, both the energy cost of running (47) and running biomechanics (48) are comparable between 0% and 1% treadmill inclines. Participants ran for 4 min at each velocity. The 11 km/h velocity was chosen because all runners in a

previous study ($n = 97$), including females with a wide performance range, were able to run at this velocity with $RER \leq 1.00$ (7). The 80% of $v\dot{V}O_{2max}$ condition was selected because it can theoretically be sustained for > 2 h without rest (49) and predicts 16 km running time (2). ECR (kcal/kg/km) was calculated as the average during the final minute of each trial/velocity using second-by-second $\dot{V}O_2$ and $\dot{V}CO_2$ data. Steady-state $\dot{V}O_2$ at both velocities was confirmed by a low coefficient of variation ($4.82 \pm 1.56\%$ at 11 km/h and $4.12 \pm 1.60\%$ at 80% of $v\dot{V}O_{2max}$) and the absence of systematic drift in the relationship between time and $\dot{V}O_2$ (slope: 0.002 ± 0.043 at 11 km/h; -0.0002 ± 0.047 at 80% of $v\dot{V}O_{2max}$). Substrate utilization (g/min) was estimated using updated nonprotein respiratory quotient equations (50). The energy derived from each substrate was then calculated by multiplying fat and carbohydrate usage by 9.75 and 4.07 kcal, respectively, reflecting their average energy content during moderate- to high-intensity exercise (51). The total energy expenditure was expressed in kcal/km. ECR values were allometrically adjusted for body mass. The allometric exponent was determined by transforming the equation $B = aM^b$ (52) into its logarithmic form: $\text{Log}(B) = \text{Log}(a) + b \cdot \text{Log}(M)$, where B is metabolic rate, a is a constant, M is body mass, and b is the exponent. Parallelism of regression lines between ECR and body mass was tested using an analysis of covariance (ANCOVA) by sex and showed no sex effect ($P = 0.608\text{--}0.755$). Power functions provided a better fit than linear models based on R^2 and root mean square error. Accordingly, the allometric exponent was obtained for all participants as $b = 0.915$ (kcal/kg^{0.915}/km) for 11 km/h and $b = 0.839$ (kcal/kg^{0.839}/km) for 80% of $v\dot{V}O_{2max}$.

Running kinematics. Thirty-two reflective markers were placed on anatomical landmarks.

Three-dimensional kinematic data during running at 11 km/h and 80% of $v\dot{V}O_{2max}$ were collected using a 21-camera motion capture system operating at 250 Hz (MAC3D, Motion Analysis

Corporation, California, USA) and Cortex software (Motion Analysis Corporation, Santa Rosa, CA). Marker trajectories were filtered with a bidirectional fourth-order low-pass Butterworth filter (cutoff frequency: 15 Hz) to remove high-frequency noise (53). Ten consecutive strides, recorded three minutes after the onset of running, were analyzed. Touch-down and toe-off were identified using a two-step procedure. First, analysis windows were defined using kinematic criteria. Touch-down was initially identified as the instant when the anterior-posterior velocity of the heel marker changed from positive to negative, ending at the subsequent minimum in vertical heel position. Toe-off was initially identified as the point when the vertical toe position (midpoint between the first and fifth metatarsal heads) exceeded 0.1 m or reached its maximum height (7, 54-56). Second, the exact instants of touch-down and toe-off were determined within these windows using acceleration-based measures. The onset of touchdown was identified by the first peak in vertical acceleration of either the heel or first metatarsal head markers, accommodating different foot strike patterns and footwear (55, 56). Toe-off was identified as the time point of peak vertical jerk (first derivative of acceleration) of the toe marker.

Kinematic variables averaged over ten steps for the right leg at each running velocity. If a marker became detached during running, its position was estimated using linear interpolation. Foot strike angle in the sagittal plane was determined from the initial joint angle at touchdown (the sagittal foot-ground angle), and treated as a continuous variable, classified as relatively dorsiflexed or plantarflexed, without categorizing runners as rearfoot or forefoot strikers, in accordance with a previous study (57). KE/KF (the sagittal thigh-shank angle) ROM was calculated as the difference between the maximum and minimum knee joint angles during the stride. In addition, mean KE/KF angular velocities during the stance phase at both 11 km/h and 80% of $\dot{V}O_{2\max}$ were calculated.

Statistical analysis

All statistical analyses were conducted using RStudio (version 2025.09.2+418, Boston, MA).

Principal component analysis. Muscle strength and jump performance variables were summarized using principal component analysis (PCA). As a result, four principal components (PCs) were extracted for each sex, which are described in detail in the Results section.

K-means clustering. Foot strike angle on the sagittal plane and KE/KF ROM were standardized into Z-scores for each sex. The ECR, the foot-strike angle, and KE/KF ROM at 80% of $v\dot{V}O_{2\max}$ were adjusted by individual running velocity using ANCOVA. The k-means clustering method was applied separately for each sex and each running velocity, using these two kinematic variables (foot strike angle and KE/KF ROM). Although multicollinearity is generally less critical in k-means clustering, we nonetheless verified that the kinematic variables used for clustering were not strongly correlated ($r = -0.164$ to 0.115). Accordingly, no additional control for multicollinearity was required. The optimal number of clusters was determined using the Elbow method: a linear reference line was drawn between cluster numbers 1 and 10, and the perpendicular distance from this line was calculated (Figure 1 (58)). The number of clusters corresponding to the greatest perpendicular distance was selected. After clustering, outliers were removed using the Distance-Based Algorithm (59) (male: $n = 6$, female: $n = 3$ at 11 km/h; male: $n = 3$, female: $n = 2$ at 80% of $v\dot{V}O_{2\max}$).

Correlation analyses. Normality was assessed using Shapiro-Wilk's test. When a variable did not meet normality assumptions, Spearman's rank correlation was used to calculate relationships between MTU functions (KE/KF torques and jump performances) and running performance-related variables (ECR and WA score) within each cluster. Correlation coefficients were

interpreted as weak ($0.0 < |r| < 0.25$), fair ($0.25 \leq |r| < 0.50$), moderate ($0.50 \leq |r| < 0.75$), or strong ($0.75 \leq |r| < 1.00$) (60). Relationships with $|r| \geq 0.5$ (moderate or strong) were considered meaningful (61). To test whether the cluster strengthened these relationships compared with the non-clustered (whole) cohort, correlation coefficients were compared using Williams' test (62).

Regression analysis. Stepwise regression analyses were performed, entering MTU function variables (PC1–4) as independent variables, and ECR or WA score as dependent variables in each cluster. In addition, the relative contribution of each retained PC was quantified using the Lindeman-Merenda-Gold method. To examine whether the relationships between MTU function and outcomes differed between clustered and non-clustered runners, linear regression slopes within each cluster were statistically compared with those obtained from the whole cohort (non-clustered runners) using interaction terms.

RESULTS

PCA

In male runners, PCA identified four components explaining 88.9% of the total variance in MTU function variables (PC1: 53.2%, PC2: 22.3%, PC3: 9.3%, PC4: 4.2%). PC1 loaded strongly on jump performance and moderately on muscle strength, representing overall MTU function (Strength & Jump [–]). PC2 was characterized by positive loadings on muscle strength and negative loadings on SSC ability (Strength [–]/SSC [–]). PC3 reflected a balance between knee extensor and flexor function, with negative loadings for KE and positive loadings for KF (KE [–]/KF [+]). PC4 loaded positively on CMJ height and negatively on jump indices, indicating non-reactive versus SSC-based jump performance (CMJ [–]/SSC ability [–]) (Table 2).

In female runners, PCA identified four components explaining 89.2% of the total variance (PC1: 48.2%, PC2: 25.2%, PC3: 10.1%, PC4: 5.8%). PC1 loaded negatively on both muscle strength and jump performance, reflecting overall MTU function, with lower scores indicating greater MTU function capacity (Strength & Jump [-]). PC2 represented a continuum from strength-dominant to jump/SSC-dominant characteristics, with positive loadings on isokinetic knee torques and negative loadings on jump performance (Strength [+]/Jump [-]). PC3 reflected knee extensor versus flexor dominance (KE [+]/KF [-]). PC4 contrasted non-reactive jump performance with reactive SSC utilization, loading positively on CMJ height and knee flexor torque and negatively on reactive jump indices and knee extensor torque (CMJ + KF [+]/SSC + KE [-]) (Table 3).

Male clusters at 11 km/h

Participants were divided into four clusters in the 11 km/h condition (Figure 2). In the non-clustered (whole-cohort) analyses, only weak to fair relationships were observed between MTU function variables (KE/KF torque and jump performance) and running performance-related variables (ECR and WA score) (ECR, $r = -0.204$ to 0.162 ; WA, $r = -0.269$ to -0.001 ; see “ALL” columns in Supplemental Table 2 Content, Supplemental Digital Content, <http://links.lww.com/MSS/D402>).

Cluster 1: dorsiflexed landing. PC4 (higher SSC ability) was selected as an associated factor with lower ECR, although this model did not reach statistical significance (adjusted $R^2 = 0.142$, $P = 0.101$). In contrast, PC1 (higher overall MTU function), together with PC4 (higher SSC ability), significantly explained the variance in WA score (adjusted $R^2 = 0.284$, $P = 0.044$).

Cluster 2: large KE/KF ROM. PC2 (higher muscle strength) and PC4 (higher SSC ability) were selected as explanatory variables for lower ECR, but the model did not reach statistical significance (adjusted $R^2 = 0.258$, $P = 0.124$).

Cluster 3: plantarflexed landing. No variables were selected to explain variance in ECR or WA score by the regression analysis.

Cluster 4: small KE/KF ROM. No variables were selected to explain variance in ECR or WA score by the regression analysis.

Female clusters at 11 km/h

Participants were divided into four clusters in the 11 km/h condition (Figure 2). In the non-clustered analyses, only weak to fair relationships were observed between MTU function variables and running performance-related variables (ECR, $r = -0.109$ to 0.267 ; WA, $r = -0.276$ to 0.175 ; see “ALL” columns in Supplemental Table 3, Supplemental Digital Content, <http://links.lww.com/MSS/D402>).

Cluster 1: dorsiflexed landing and large KE/KF ROM. PC2 (higher jump performance) significantly explained the variance in both ECR (adjusted $R^2 = 0.256$, $P = 0.013$) and WA (adjusted $R^2 = 0.259$, $P = 0.013$).

Cluster 2: plantarflexed landing and large KE/KF ROM. PC2 (higher jump performance) and PC4 (higher CMJ + KF torques) were selected as explanatory variables for lower ECR; however, these associations did not reach statistical significance (adjusted $R^2 = 0.285$, $P = 0.128$). In addition, the linear regression slope of PC4 was significantly different from that of the non-clustered runners ($P = 0.008$).

Cluster 3: plantarflexed landing and small KE/KF ROM. PC1 (higher overall MTU function) and PC4 (higher CMJ height and KF torques) were selected as explanatory variables for lower

ECR, although the model was not statistically significant ($\text{adjusted } R^2 = 0.214, P = 0.324$).

Similarly, PC2 (higher jump performance) and PC3 (higher KE torque) were associated with the variance in WA score, but this relationship did not reach statistical significance ($\text{adjusted } R^2 = 0.667, P = 0.089$).

Cluster 4: dorsiflexed landing and small KE/KF ROM. PC1 (higher CMJ height and KF torques), PC2 (higher jump performance), and PC4 (higher CMJ + KF torque) were significantly associated with lower ECR ($\text{adjusted } R^2 = 0.356, P = 0.015$). PC1 (lower overall MTU function) and PC2 (higher jump performance) contributed to the variance in WA score, indicating that jump performance rather than muscle strength was associated with WA score, while this association did not reach statistical significance ($\text{adjusted } R^2 = 0.164, P = 0.078$).

Male clusters at 80% of $\dot{V}O_{2\max}$

Participants were divided into four clusters in the 80% $\dot{V}O_{2\max}$ condition (Figure 3). In the non-clustered analyses, only weak to fair relationships were observed between MTU function variables and running performance-related variables (ECR, $r = -0.034$ to 0.115 ; see “ALL” columns in Supplemental Table 4, Supplemental Digital Content, <http://links.lww.com/MSS/D402>).

Cluster 1: dorsiflexed landing and large KE/KF ROM. PC2 (higher SSC ability) significantly explained the variance in ECR ($\text{adjusted } R^2 = 0.483, P = 0.007$), and its regression slope was significantly different from that of non-clustered runners ($P = 0.016$). In addition, PC1 (lower overall MTU function) and PC2 (higher strength) were significantly associated with WA score ($\text{adjusted } R^2 = 0.553, P = 0.011$), indicating that muscle strength rather than jump performance was associated with WA score. The linear regression slope of PC1 significantly differed from that of non-clustered runners ($P = 0.036$).

Cluster 2: plantarflexed landing and large KE/KF ROM. PC4 (higher CMJ height) was associated with WA score, although this relationship did not reach statistical significance (adjusted $R^2 = 0.079$, $P = 0.172$).

Cluster 3: plantarflexed landing and small KE/KF ROM. PC4 (higher SSC ability) significantly explained the variance in both ECR (adjusted $R^2 = 0.358$, $P = 0.031$) and WA score (adjusted $R^2 = 0.383$, $P = 0.025$).

Cluster 4: dorsiflexed landing and small KE/KF ROM. PC1 (higher overall MTU function) was selected as an explanatory variable for ECR, while it was not significant (adjusted $R^2 = 0.056$, $P = 0.161$). In addition, PC1 (higher overall MTU function) and PC4 (higher SSC ability) were associated with WA score (adjusted $R^2 = 0.098$, $P = 0.162$), although these relationships did not reach statistical significance. The linear regression slope of this model significantly differed from that of the non-clustered cohort ($P = 0.036$).

Female clusters at 80% of $\dot{V}O_{2\max}$

Participants were divided into three clusters in the 80% $\dot{V}O_{2\max}$ condition (Figure 3). In the non-clustered analyses, only weak to fair relationships were observed between MTU function variables and running performance-related variables (ECR, $r = -0.137$ to 0.224 ; see “ALL” columns in Supplemental Table 5, Supplemental Digital Content, <http://links.lww.com/MSS/D402>).

Cluster 1: dorsiflexed landing. PC1 (higher overall MTU function) and PC4 (higher CMJ height and KF torque) were significantly associated with ECR (adjusted $R^2 = 0.352$, $P = 0.038$). In addition, PC2 (higher jump performance), PC3 (higher KE torque), and PC4 (higher CMJ height and KF torque) significantly explained the variance in WA score (adjusted $R^2 = 0.595$, $P = 0.016$).

Cluster 2: plantarflexed landing and large KE/KF ROM. PC2 (higher jump performance) was selected as an explanatory factor for ECR, although this relationship did not reach statistical significance (adjusted $R^2 = 0.057$, $P = 0.153$).

Cluster 3: plantarflexed landing and small KE/KF ROM. PC1 (lower overall MTU function) and PC2 (higher jump performance) significantly explained the variance in ECR (adjusted $R^2 = 0.159$, $P = 0.047$), indicating that jump performance rather than muscle strength was associated with ECR. In addition, PC2 (higher jump performance) and PC4 (higher CMJ height and KF torque) were significantly associated with WA score (adjusted $R^2 = 0.222$, $P = 0.024$), and the linear regression slope was significantly different from that of the non-clustered cohort ($P = 0.015$).

DISCUSSION

Overall shared trends

In both sexes, whole-cohort analyses revealed only weak to fair correlations between MTU function and running economy and/or performance at both 11 km/h and 80% of $v\dot{V}O_{2max}$. Several of these trends were significantly strengthened when runners were stratified into kinematics-based clusters, with the patterns varying by cluster. This finding supports our first hypothesis that clustering based on foot strike angle and KE/KF ROM can enhance the detection of associations between MTU function and performance-related outcomes. However, most clusters did not consistently align with the relationships hypothesized in the Background (i.e., dorsiflexed landing with KF function; large KE/KF ROM with KE function; and plantarflexed landing and small KE/KF ROM with jump performance), and these associations were not uniformly reinforced. Specifically, in male runners, significant explanatory variables accounted for 1 out of 8 clusters (12.5%) at 11 km/h, whereas 3 out of 8 clusters (37.5%) were significant at

80% of $\dot{V}O_{2\max}$. In female runners, 4 out of 8 clusters (50.0%) reached significance at 11 km/h and 4 out of 6 clusters (66.7%) at 80% of $\dot{V}O_{2\max}$. Accordingly, we interpret the findings by first addressing the overall trends and then considering the moderating effects of sex and running velocity.

Unexpectedly, a consistent pattern was observed across sexes and running conditions, whereby jump performance played a more important role in running economy and performance in dorsiflexed clusters than in plantarflexed clusters. This may be attributable to the greater braking impulse experienced by rearfoot strikers during the early stance phase compared with forefoot strikers (63), which can negatively affect running economy (7). Indeed, rearfoot strikers (i.e., dorsiflexed runners) have been shown to exhibit greater activation of knee-joint musculature during the early stance phase to attenuate the larger braking impulse, compared with forefoot strikers (63). Accordingly, reducing the braking phase and enhancing the efficient storage and release of elastic energy through SSC function may be particularly beneficial for rearfoot strikers. In contrast, forefoot strikers generally exhibit greater activation and co-activation of muscles surrounding the ankle joint rather than the knee joint during running (63, 64). In the present study, drop and rebound jumps may partially reflect ankle-joint musculotendinous SSC ability; however, these tasks are also characterized by a substantial contribution of knee-joint-dominant SSC function (65-67). Taken together, these findings suggest that jump performance, particularly knee-dominant musculotendinous SSC ability, may represent a limiting factor for running economy and performance in dorsiflexed runners.

Sex differences

In this section, we focus on three key aspects of the results in female runners. First, jump performance emerged as an important determinant of both running economy and performance

across multiple female clusters. Second, the overall MTU function was particularly relevant in female runners characterized by small KE/KF ROM. Third, a relatively larger number of MTU-related variables were selected in the regression models for female clusters compared with males. Together, these findings suggest a more complex and heterogeneous contribution of MTU function to running economy and performance in female runners, which is discussed below.

Firstly, although we expected that KE torque would be associated with running economy and performance in female runners, jump performance emerged as a key factor associated with both running economy and performance in most female clusters. Sex-specific neuromuscular control strategies may partly explain this finding. Previous studies have reported that females exhibit a higher frequency of muscle co-contraction during walking (68, 69) and jumping tasks (70) than males, which increases the demand for neuromuscular capacity during SSC actions. In addition, adult females tend to preferentially activate the rectus femoris to stabilize the knee joint, as this biarticular muscle contributes not only to KE torque production but also to the control of hip excursion during landing and jumping (71-73). In the present study, PCA in female runners also revealed a co-variation between drop and rebound jump performance and KE torque, particularly slow KE torque, suggesting that KE function may indirectly contribute to superior running economy and performance through its association with jump-related SSC performance. Collectively, these findings indicate that female runners may rely on a neuromuscular strategy characterized by enhanced co-contraction and quadriceps-dominant control during running, thereby increasing the importance of jump performance for running economy and performance.

Secondly, in female runners, most clusters characterized by small KE/KF ROM showed the importance of overall MTU function for improved running economy and performance. In

contrast, in male runners with small KE/KF ROM, SSC ability was selectively associated with running economy and performance, particularly in Cluster 3 at 80% of $\dot{V}O_{2\max}$. This finding suggests that greater tendon stiffness in males may facilitate rapid force transmission and modulation of muscle activation during short ground contact times (24, 74). In contrast, female runners exhibited a broader dependence on MTU function, encompassing both muscle strength and jump performance. As mentioned above, females tend to rely more on muscle co-contraction to enhance joint stability (68-70) and compensate for lower tendon stiffness (74). However, this strategy may limit effective utilization of the SSC by increasing muscle fascicle work and reducing the amount of elastic energy stored and returned by tendinous tissues (75, 76). Consequently, greater muscle strength is required to maintain leg stiffness, while superior jump performance becomes necessary to offset reduced SSC efficiency. Together, these findings suggest that male runners with small KE/KF ROM can benefit more selectively from SSC ability through effective force transmission, whereas female runners with small KE/KF ROM require a wider range of MTU functions to maintain running economy and performance.

Thirdly, a relatively greater number of MTU-related variables were selected for female clusters compared with males at 11 km/h. At a given absolute running speed, female runners typically run at a higher intensity than male runners (77). Although direct evidence remains limited, this higher relative intensity may amplify the contribution of MTU function, including both muscle strength and SSC-related performance, to running economy and performance in female runners. Indeed, available data suggest that running at higher relative intensities increases reliance on muscle force production and reactive strength, irrespective of sex (5). In addition, females may depend more on active muscle force to meet mechanical demands under conditions of higher relative intensity (78), as sex differences in neuromuscular activation and fatigability

have been reported (79). Consequently, the greater number of MTU-related variables selected in female clusters may reflect not only sex-specific neuromuscular strategies but also the higher relative intensity at which female runners performed the running tasks, underscoring the importance of overall MTU function for maintaining running economy and performance. Nevertheless, the number of studies directly examining sex-specific determinants of running economy and performance across different relative intensities remains limited, highlighting a substantial need for further research in this area.

Effects of running velocities

Our hypothesis that the relationships of MTU function with running economy and performance would be strengthened in a cluster-specific manner was grounded in prior studies conducted at the same absolute running velocity across runners (28, 29, 34-36), because few studies have examined running kinematics or profiles at a matched relative velocity. Nevertheless, a greater number of MTU-related variables were associated with running economy and/or performance at 80% of $\dot{V}O_{2\max}$ than at 11 km/h, regardless of sex, with this tendency being particularly evident in male runners. Importantly, these patterns remained after statistical adjustment for running velocity when examining the relationships between joint angles and ECR. This adjustment underscores the utility of accounting for running velocity when evaluating MTU-related determinants under relative intensity conditions.

In male runners, a clear velocity-dependent shift was observed in the associations between muscle–tendon unit function and running economy and performance, with substantially more significant relationships emerging at 80% of $\dot{V}O_{2\max}$ than at 11 km/h. At the lower speed, the mechanical demands required to generate vertical and propulsive impulse appear to be relatively modest (16, 78), thereby possibly reducing the sensitivity of running economy and

performance to inter-individual differences in MTU function. As running velocity increases, however, contact time shortens, and the temporal constraints on impulse generation become more stringent (16, 78), requiring more rapid force development and more effective jump performance (Figure 3). Consequently, impulse generation becomes increasingly dependent on active musculotendinous function, a pattern likely to be particularly pronounced in male runners for whom the difference between 11 km/h and 80% of $v\dot{V}O_{2max}$ represents a substantial increase in relative and absolute mechanical demand. This interpretation is consistent with biomechanical evidence indicating that MTU functions become critical determinants of running economy and performance at higher running intensities.

In contrast, female runners showed no marked velocity-dependent change in the overall pattern of associations, suggesting that the fundamental muscular demands required to generate the necessary impulse were already present at both running velocities. This finding may reflect a neuromuscular strategy in which impulse generation relies more consistently on MTU functions across speeds, likely due to a greater need for muscular stabilization (71-73) and lower passive tendon stiffness (74). As a result, increasing running velocity did not introduce new limiting factors in terms of overall MTU function. Nevertheless, compared with the 11 km/h condition, the importance of CMJ performance and KF torque in relation to ECR increased at 80% of $v\dot{V}O_{2max}$. Although the increase in running velocity was relatively modest in female runners (absolute, 11.0 km/h; relative, 12.4 ± 0.9 km/h), previous evidence indicates that knee flexor activation increases with running speed, even with a slight running velocity change (9.4–13.3 km/h) (80). Moreover, females have been reported to exhibit a greater involvement of the KF activation during CMJ compared with males (81). Accordingly, even without a marked increase in the overall reliance on MTU function across speeds, relatively higher-intensity running in

females may selectively amplify the contribution of knee flexor function and, secondarily, CMJ-related SSC capacity to efficient force production and braking control. This highlights sex-specific differences in how velocity-related mechanical demands are accommodated.

Limitations

This study has some limitations. First, clusters were derived from the relative positions of runners within our sample of 120 Japanese endurance runners; thus, generalizability to other ages, populations, and competition levels remains uncertain. Preliminary longitudinal observations in our cohort suggest that individual cluster membership is relatively stable over time (unpublished data), supporting the use of this approach to identify performance-relevant MTU function in similar populations. Future multi-center data pooling will be important for external validation and for developing individualized, kinematics-informed training prescriptions tailored to sex, competition level, and population background. Second, the cross-sectional design precludes causal inference. Whether targeted strength training improves running economy and performance in a cluster-specific manner remains to be determined. Randomized, cluster-stratified training interventions are needed to establish causality and to evaluate potential adverse responses in specific kinematic profiles. Third, the timing of data collection was not standardized across participants, as measurements were conducted across different phases of the training season (e.g., off-season and in-season) (82). Running economy, muscle strength, and jump performance may fluctuate over a periodized training cycle. Although significant relationships between MTU-related variables and ECR were observed, the magnitude of these associations may differ depending on seasonal training status. Therefore, future longitudinal studies with controlled measurement timing are needed.

CONCLUSIONS

Classifying endurance runners by foot strike angle (sagittal plane) and KE/KF ROM strengthened several relationships between MTU function (strength and jump performance) and running economy/performance compared with whole-cohort analyses, and did so in a cluster-specific manner, especially at 80% of $\dot{V}O_{2\max}$. Specifically, jump performance correlated with running economy and performance in clusters with dorsiflexed landing and female runners overall, whereas overall MTU function was associated with running economy and performance in clusters characterized by small KE/KF ROM. Notably, female runners exhibited a broader dependence on MTU-related variables, suggesting a more complex and heterogeneous contribution of muscle strength and SSC-related capacity to running economy and performance compared with males. These findings support kinematics-based clustering as a practical framework for individualizing strength training prescriptions to enhance running economy and performance, while accounting for sex-specific and kinematics-dependent variations.

REFERENCES

1. The state of running 2019 [Internet]. RunRepeat; 2024 [cited 2026 Feb 10]. Available from: <https://runrepeat.com/state-of-running>.
2. McLaughlin JE, Howley ET, Bassett DR, Thompson DL, Fitzhugh EC. Test of the classic model for predicting endurance running performance. *Med Sci Sports Exerc.* 2010;42(5):991-7.
3. Midgley AW, McNaughton LR, Wilkinson M. Is there an optimal training intensity for enhancing the maximal oxygen uptake of distance runners? *Sports Med.* 2006;36(2):117-32.
4. Shaw AJ, Ingham SA, Folland JP. The valid measurement of running economy in runners. *Med Sci Sports Exerc.* 2014;46(10):1968-73.
5. Li F, Newton RU, Shi Y, Sutton D, Ding H. Correlation of eccentric strength, reactive strength, and leg stiffness with running economy in well-trained distance runners. *J Strength Cond Res.* 2021;35(6):1491-9.
6. Zhang Q, Nassis GP, Chen S, Shi Y, Li F. Not lower-limb joint strength and stiffness but vertical stiffness and isometric force-time characteristics correlate with running economy in recreational male runners. *Front Physiol.* 2022;13:940761.
7. Folland JP, Allen SJ, Black MI, Handsaker JC, Forrester SE. Running technique is an important component of running economy and performance. *Med Sci Sports Exerc.* 2017;49(7):1412-23.
8. Pizzuto F, de Oliveira CF, Soares TSA, Rago V, Silva G, Oliveira J. Relationship between running economy and kinematic parameters in long-distance runners. *J Strength Cond Res.* 2019;33(7):1921-8.
9. Trowell D, Phillips E, Saunders PU, Bonacci J. The relationship between performance and biomechanics in middle-distance runners. *Sports Biomech.* 2021;20(8):974-84.

10. Saunders PU, Telford RD, Pyne DB, et al. Improved running economy in elite runners after 20 days of simulated moderate-altitude exposure. *J Appl Physiol* (1985). 2003;96(3):931-7.
11. Hunter GR, McCarthy JP, Carter SJ, et al. Muscle fiber type, Achilles tendon length, potentiation, and running economy. *J Strength Cond Res*. 2015;29(5):1302-9.
12. Bohm S, Mersmann F, Santuz A, Schroll A, Arampatzis A. Muscle-specific economy of force generation and efficiency of work production during human running. *Elife*. 2021;10:e67182.
13. Blagrove RC, Howe LP, Cushion EJ, et al. Effects of strength training on postpubertal adolescent distance runners. *Med Sci Sports Exerc*. 2018;50(6):1224-32.
14. Eihara Y, Takao K, Sugiyama T, et al. Heavy resistance training versus plyometric training for improving running economy and running time trial performance: a systematic review and meta-analysis. *Sports Med Open*. 2022;8(1):138.
15. Barnes KR, Kilding AE. Strategies to improve running economy. *Sports Med*. 2015;45(1):37-56.
16. Schache AG, Blanch PD, Dorn TW, Brown NA, Rosemond D, Pandy MG. Effect of running speed on lower limb joint kinetics. *Med Sci Sports Exerc*. 2011;43(7):1260-71.
17. Sasaki K, Neptune RR. Muscle mechanical work and elastic energy utilization during walking and running near the preferred gait transition speed. *Gait Posture*. 2006;23(3):383-90.
18. Morin JB, Gimenez P, Edouard P, et al. Sprint acceleration mechanics: the major role of hamstrings in horizontal force production. *Front Physiol*. 2015;6:404.
19. Dorn TW, Schache AG, Pandy MG. Muscular strategy shift in human running: dependence of running speed on hip and ankle muscle performance. *J Exp Biol*. 2012;215(Pt 11):1944-56.

20. Tam N, Santos-Concejero J, Coetzee DR, Noakes TD, Tucker R. Muscle co-activation and its influence on running performance and risk of injury in elite Kenyan runners. *J Sports Sci.* 2017;35(2):175-81.
21. Chen S, Van Hooren B, Jiang S, et al. Interaction between muscle activation, co-activation, and lower extremity stiffness in recreational runners: a statistical parametric mapping approach. *Med Sci Sports Exerc.* 2025;57(11):2419-2428.
22. Van Hooren B, Jukic I, Cox M, Frenken KG, Bautista I, Moore IS. The relationship between running biomechanics and running economy: a systematic review and meta-analysis of observational studies. *Sports Med.* 2024;54(5):1269-316.
23. Kyröläinen H, Finni T, Avela J, Komi PV. Neuromuscular behaviour of the triceps surae muscle-tendon complex during running and jumping. *Int J Sports Med.* 2003;24(3):153-5.
24. Bohm S, Mersmann F, Santuz A, Arampatzis A. Enthalpy efficiency of the soleus muscle contributes to improvements in running economy. *Proc Biol Sci.* 2021;288(1943):20202784.
25. Festa L, Tarperi C, Skroce K, et al. Effects of flywheel strength training on the running economy of recreational endurance runners. *J Strength Cond Res.* 2019;33(3):684-90.
26. Li F, Wang R, Newton RU, Sutton D, Shi Y, Ding H. Effects of complex training versus heavy resistance training on neuromuscular adaptation, running economy and 5-km performance in well-trained distance runners. *PeerJ.* 2019;7:e6787.
27. Patoz A, Breine B, Thouvenot A, Mourot L, Gindre C, Lussiana T. Does characterizing global running pattern help to prescribe individualized strength training in recreational runners? *Front Physiol.* 2021;12:631637.
28. Chumanov ES, Wall-Scheffler C, Heiderscheit BC. Gender differences in walking and running on level and inclined surfaces. *Clin Biomech (Bristol).* 2008;23(10):1260-8.

29. Malinzak RA, Colby SM, Kirkendall DT, Yu B, Garrett WE. A comparison of knee joint motion patterns between men and women in selected athletic tasks. *Clin Biomech (Bristol)*. 2001;16(5):438-45.
30. Santuz A, Ekizos A, Janshen L, et al. Modular control of human movement during running: an open access data set. *Front Physiol*. 2018;9:1509.
31. Shultz SJ, Beynnon BD, Schmitz RJ. Sex differences in coupled knee motions during the transition from non-weight bearing to weight bearing. *J Orthop Res*. 2009;27(6):717-23.
32. Schmitz RJ, Shultz SJ. Contribution of knee flexor and extensor strength on sex-specific energy absorption and torsional joint stiffness during drop jumping. *J Athl Train*. 2010;45(5):445-52.
33. Nakagawa K, Takamatsu R, Matsushima M. Sex differences in muscle activity during drop-jump landing motion. *J Health Sci Med*. 2022;5(3):28-35.
34. Yong JR, Silder A, Delp SL. Differences in muscle activity between natural forefoot and rearfoot strikers during running. *J Biomech*. 2014;47(15):3593-7.
35. Lussiana T, Gindre C, Hebert-Losier K, Sagawa Y, Gimenez P, Mourot L. Similar running economy with different running patterns along the aerial-terrestrial continuum. *Int J Sports Physiol Perform*. 2017;12(4):481-9.
36. Ervilha UF, Mochizuki L, Figueira A Jr, Hamill J. Are muscle activation patterns altered during shod and barefoot running with a forefoot footfall pattern? *J Sports Sci*. 2017;35(17):1697-703.
37. Kusagawa Y, Kurihara T, Maeo S, Sugiyama T, Kanehisa H, Isaka T. A classification of the plantar intrinsic foot muscles based on the physiological cross-sectional area and muscle fiber length in healthy young adult males. *J Foot Ankle Res*. 2023;16(1):75.

38. Suyal M, Sharma S. A review on analysis of k-means clustering machine learning algorithm based on unsupervised learning. *J Artif Intell.* 2024;6(1):85-95.
39. McKay AKA, Stellingwerff T, Smith ES, et al. Defining training and performance caliber: A participant classification framework. *Int J Sports Physiol Perform.* 2022;17(2):317-31.
40. World Athletics scoring tables updated for 2025. [cited 2026 Feb 10]; Available from: <https://worldathletics.org/news/news/scoring-tables-2025>.
41. Manole LM, Ghiga G, Iftinchi O, et al. Bioelectrical impedance analysis versus dual x-ray absorptiometry for obesity assessment in pediatric populations: a systematic review. *Diagnostics.* 2025;15(12):1505.
42. Looney DP, Schafer EA, Chapman CL, et al. Reliability, biological variability, and accuracy of multi-frequency bioelectrical impedance analysis for measuring body composition components. *Front Nutr.* 2024;11:1491931.
43. Glatthorn JF, Gouge S, Nussbaumer S, Stauffacher S, Impellizzeri FM, Maffiuletti NA. Validity and reliability of Optojump photoelectric cells for estimating vertical jump height. *J Strength Cond Res.* 2011;25(2):556-60.
44. Comyns TM, Murphy J, O'Leary D. Reliability, usefulness, and validity of field-based vertical jump measuring devices. *J Strength Cond Res.* 2023;37(8):1594-9.
45. Howley ET, Bassett DR Jr, Welch HG. Criteria for maximal oxygen uptake: review and commentary. *Med Sci Sports Exerc.* 1995;27(9):1292-301.
46. Damasceno MV, Lima-Silva AE, Pasqua LA, et al. Effects of resistance training on neuromuscular characteristics and pacing during 10-km running time trial. *Eur J Appl Physiol.* 2015;115(7):1513-22.

47. Miller JR, Van Hooren B, Bishop C, Buckley JD, Willy RW, Fuller JT. A systematic review and meta-analysis of crossover studies comparing physiological, perceptual and performance measures between treadmill and overground running. *Sports Med.* 2019;49(5):763-82.
48. Van Hooren B, Fuller JT, Buckley JD, et al. Is motorized treadmill running biomechanically comparable to overground running? A systematic review and meta-analysis of cross-over studies. *Sports Med.* 2019;50(4):785-813.
49. Hoogkamer W, Kram R, Arellano CJ. How biomechanical improvements in running economy could break the 2-hour marathon barrier. *Sports Med.* 2017;47(9):1739-50.
50. Péronnet F, Massicotte D. Table of nonprotein respiratory quotient: an update. *Can J Sport Sci.* 1991;16(1):23-9.
51. Jeukendrup AE, Wallis GA. Measurement of substrate oxidation during exercise by means of gas exchange measurements. *Int J Sports Med.* 2005;26:S28-S37.
52. Huxley JS. Constant differential growth-ratios and their significance. *Nature.* 1924;114:895-6.
53. Winter DA. Processing of raw kinematic data: smoothing and fitting of data. 3rd ed. Hoboken (NJ): John Wiley & Sons; 1990. p.42-50
54. Alvim F, Cerqueira L, Netto AD, Leite G, Muniz A. Comparison of five kinematic-based identification methods of foot contact events during treadmill walking and running at different speeds. *J Appl Biomech.* 2015;31(5):383-8.
55. Maiwald C, Sterzing T, Mayer TA, Milani TL. Detecting foot-to-ground contact from kinematic data in running. *Footwear Sci.* 2009;1(2):111-8.

56. Handsaker JC, Forrester SE, Folland JP, Black MI, Allen SJ. A kinematic algorithm to identify gait events during running at different speeds and with different footstrike types. *J Biomech.* 2016;49(16):4128-33.
57. Altman AR, Davis IS. A kinematic method for footstrike pattern detection in barefoot and shod runners. *Gait Posture.* 2012;35(2):298-300.
58. Delgado H, Anguera X, Fredouille C, Serrano J. Novel clustering selection criterion for fast binary key speaker diarization. INTERSPEECH; Dresden, Germany, 2015. p.3091-5.
59. Saha PK, Logofatu D. Efficient approaches for density-based spatial clustering of applications with noise. In: Maglogiannis, I., Macintyre, J., Iliadis, L, editors. IFIP Advances in Information and Communication Technology, vol 627. Champain: Springer; 2021. p.184-95.
60. Batterham AM, Hopkins WG. Making meaningful inferences about magnitudes. *Int J Sports Physiol Perform.* 2006;1(1):50-7.
61. Blessie EC, Karthikeyan E. Sigmis: A feature selection algorithm using correlation based method. *J Algorithms Comput Technol.* 2012;6(3):385-94.
62. Olivier J, Diedenhofen B, Musch J. cocor: A Comprehensive solution for the statistical comparison of correlations. *PLoS One.* 2015;10(4):e0131499.
63. Xu Y, Yuan P, Wang R, Wang D, Liu J, Zhou H. Effects of foot strike techniques on running biomechanics: a systematic review and meta-analysis. *Sports Health.* 2021;13(1):71-7.
64. Chen H, Sun D, Zhang Q, Liu Q, Bíró I, Gu Y. Biomechanical differences between habitual forefoot and rearfoot strike running: a systematic review. *Adv Exerc Health Sci.* 2024;1(4):241-7.
65. Bobbert MF, Huijing PA, van Ingen Schenau GJ. Drop jumping. I. The influence of jumping technique on the biomechanics of jumping. *Med Sci Sports Exerc.* 1987;19(4):332-8.

66. Cè E, Zushi A, Yoshida T, Zushi K, Kariyama Y, Ogata M. Characteristics of three lower limb joint kinetics affecting rebound jump performance. *PLoS One*. 2022;17(8):e0268339.
67. Xu J, Turner A, Comyns TM, Harry JR, Chavda S, Bishop C. Countermovement rebound jump: a comparison of joint work and joint contribution to the countermovement and drop jump tests. *Appl Sci*. 2023;13(19):10680.
68. Mengarelli A, Maranesi E, Barone V, Burattini L, Fioretti S, Di Nardo F. Evaluation of gender-related differences in co-contraction activity of shank muscles during gait. *Annu Int Conf IEEE Eng Med Biol Soc*. 2015;2015:6066-9.
69. Strazza A, Mengarelli A, Verdini F, et al. Increased co-contraction activity during push-off phase of walking in healthy women. *IRBM*. 2021;42(1):48-54.
70. Otsuki R, Del Bel MJ, Benoit DL. Sex differences in muscle activation patterns associated with anterior cruciate ligament injury during landing and cutting tasks: a systematic review. *J Electromyogr Kinesiol*. 2021;60:102583.
71. Hewett TE, Stroupe AL, Nance TA, Noyes FR. Plyometric training in female athletes. Decreased impact forces and increased hamstring torques. *Am J Sports Med*. 1996;24(6):765-73.
72. Youdas JW, Hollman JH, Hitchcock JR, Hoyme GJ, Johnsen JJ. Comparison of hamstring and quadriceps femoris electromyographic activity between men and women during a single-limb squat on both a stable and labile surface. *J Strength Cond Res*. 2007;21(1):105-11.
73. Hedayatpour N, Mohammed Sediq Rashid D, Izanloo Z, Seylانه H, Falla D. Men and women show different adaptations of quadriceps activity following fatiguing contractions: An explanation for the increased incidence of sports-related knee injuries in women? *J Electromyogr Kinesiol*. 2021;58:102552.

74. Taş S, Salkın Y. An investigation of the sex-related differences in the stiffness of the Achilles tendon and gastrocnemius muscle: Inter-observer reliability and inter-day repeatability and the effect of ankle joint motion. *Foot (Edinb)*. 2019;41:44-50.
75. Ishikawa M, Komi PV, Grey MJ, Lepola V, Bruggemann GP. Muscle-tendon interaction and elastic energy usage in human walking. *J Appl Physiol (1985)*. 2005;99(2):603-8.
76. Farris DJ, Sawicki GS. Human medial gastrocnemius force-velocity behavior shifts with locomotion speed and gait. *Proc Natl Acad Sci USA*. 2012;109(3):977-82.
77. Cavanagh PR, Williams KR. The effect of stride length variation on oxygen uptake during distance running. *Med Sci Sports Exerc*. 1982;14(1):30-5.
78. Jiang X, Bíró I, Sárosi J, Fang Y, Gu Y. Comparison of ground reaction forces as running speed increases between male and female runners. *Front Bioeng Biotechnol*. 2024;12:1378284.
79. Hunter SK. Sex differences in human fatigability: mechanisms and insight to physiological responses. *Acta Physiologica*. 2014;210(4):768-89.
80. Orendurff MS, Kobayashi T, Tulchin-Francis K, et al. A little bit faster: lower extremity joint kinematics and kinetics as recreational runners achieve faster speeds. *J Biomech*. 2018;71:167-75.
81. Márquez G, Alegre LM, Jaén D, Martín-Casado L, Aguado X. Sex differences in kinetic and neuromuscular control during jumping and landing. *J Musculoskelet Neuronal Interact*. 2017;17(1):409-16.
82. Saunders PU, Cox AJ, Hopkins WG, Pyne DB. Physiological measures tracking seasonal changes in peak running speed. *Int J Sports Physiol Perform*. 2010;5(2):230-8.

FIGURE LEGENDS

Figure 1. Elbow analysis for determining the optimal number of clusters. The graph shows the within-cluster sum of squares (WCSS) for cluster numbers 1, 3–5, and 10. "L" represents the length of the perpendicular line, indicating the optimal number of clusters based on the longest distance.

Figure 2. Factors associated with energy cost of running and WA score when runners are classified based on running kinematics at 11 km/h.

Foot strike angle and KE/KF ROM during the stance phase were converted into Z-scores (Left). The bar plots on the right depict the relative contributions of regression-selected variables using the Lindeman-Merenda-Gold method (Right). The sign in the bar graph indicates the regression coefficient, and the accompanying variable is associated with the outcome. For example, "[+]: SSC ability" indicates that the coefficient of PC4 is positive, and therefore, higher SSC ability is associated with the outcome.

* indicates a significant model adjusted R^2 , and # indicates a regression slope significantly different from the non-clustered analysis. 'NOT' in the interpretation of PC loading indicates that a lower value is associated with lower ECR or higher WA score.

Abbreviations: PC, principal component; SSC, stretch-shortening cycle; KE, knee extension torque; KF, knee flexion torque; CMJ, countermovement jump.

Figure 3. Factors associated with energy cost of running and WA score when runners are classified based on running kinematics at 80% of $\dot{V}O_{2max}$.

Foot strike angle and KE/KF ROM during the stance phase were converted into Z-scores (Left). The bar plots on the right depict the relative contributions of regression-selected variables using the Lindeman-Merenda-Gold method (Right). The sign in the bar graph indicates the regression

coefficient, and the accompanying variable is associated with the outcome. For example, “[−]: SSC ability” indicates that the coefficient of PC2 is negative, and therefore, higher SSC ability is associated with the outcome.

* indicates a significant model adjusted R^2 , and # indicates a regression slope significantly different from the non-clustered analysis. ‘NOT’ in the interpretation of PC loading indicates that a lower value is associated with lower ECR or higher WA score.

Abbreviations: PC, principal component; SSC, stretch-shortening cycle; KE, knee extension torque; KF, knee flexion torque; CMJ, countermovement jump.

SUPPLEMENTAL DIGITAL CONTENT

SDC 1: Supplemental Digital Content.docx

Table 1. The characteristics of participants.

	Male (n = 60)	Female (n = 60)
Age (years)	34.1 ± 10.3	34.1 ± 9.7
Height (m)	1.71 ± 0.06	1.59 ± 0.05
Body mass (kg)	60.0 ± 5.9	49.4 ± 5.9
Fat percentage (%)	11.2 ± 3.4	19.5 ± 5.3
WA score	527.7 ± 224.7	618.4 ± 183.2
Energy cost at 11 km/h (kcal/kg ^{0.915} /km)	1.48 ± 0.13	1.47 ± 0.13
Energy cost at 80% $\dot{V}O_{2max}$ (kcal/kg ^{0.839} /km)	1.96 ± 0.13	1.98 ± 0.14
$\dot{V}O_{2max}$ (mL/kg/min)	55.4 ± 5.3	46.3 ± 5.5
$\dot{V}O_{2max}$ (km/h)	18.1 ± 1.6	15.5 ± 1.1

Data are presented as mean ± SD.

The seasonal record over distances from 5000 m to the ultra-marathon (100 km) was converted to a WA score.

Table 2. Factor loadings of the principal component analysis (PCA) on MTU-related variables for males.

Variables	PC1	PC2	PC3	PC4
Isokinetic slow KE	<u>-0.152</u>	-0.509	<u>0.533</u>	0.063
Isokinetic fast KE	<u>-0.259</u>	-0.422	<u>0.427</u>	0.067
Isokinetic slow KF	<u>-0.261</u>	-0.426	0.461	-
Isokinetic fast KF	<u>-0.262</u>	-0.402	0.524	-
CMJ height	-0.368	0.170	0.140	0.801
DJ height	-0.407	0.151	0.023	0.159
DJ index	-0.399	<u>0.200</u>	0.062	<u>0.391</u>
RJ height	-0.400	<u>0.239</u>	0.115	0.058
RJ index	-0.392	<u>0.265</u>	0.095	<u>0.377</u>
Standard deviation	2.188	1.416	0.914	0.613
Proportion of variance	0.532	0.223	0.093	0.042
Cumulative proportion	0.532	0.755	0.847	0.889
Interpretation	Strength & Jump (-)	Strength (-)/SSC ability (+)	KE (-)/KF (+)	CMJ (+)/SSC ability (-)

Abbreviations: PC, principal component; KE, knee extension torque; KF, knee flexion torque;

DJ, drop jump; RJ, rebound jump; SSC, stretch-shortening cycle.

Table 3. Factor loadings of the principal component analysis (PCA) on MTU-related variables for females.

Variables	PC1	PC2	PC3	PC4
Isokinetic slow KE	-0.463	0.550	0.206	<u>-0.315</u>
Isokinetic fast KE	<u>-0.156</u>	0.392	0.728	0.080
Isokinetic slow KF	<u>-0.209</u>	0.478	<u>-0.399</u>	0.114
Isokinetic fast KF	<u>-0.256</u>	0.397	<u>0.404</u>	0.333
CMJ height	-0.343	<u>-0.230</u>	<u>0.266</u>	0.701
DJ height	-0.423	<u>-0.188</u>	0.063	-0.183
DJ index	-0.416	<u>-0.151</u>	0.016	<u>-0.478</u>
RJ height	-0.435	<u>-0.170</u>	-0.028	0.040
RJ index	-0.425	<u>-0.135</u>	-0.170	-0.123
Standard deviation	2.082	1.504	0.953	0.723
Proportion of variance	0.482	0.252	0.101	0.058
Cumulative proportion	0.482	0.733	0.834	0.892
Interpretation	Strength & Jump (-)	Strength (+)/ Jump (-)	KE (+)/ KF (-)	CMJ + KF (+)/ SSC + KE (-)

Abbreviations: PC, principal component; KE, knee extension torque; KF, knee flexion torque; DJ, drop jump; RJ, rebound jump; SSC, stretch-shortening cycle.

Figure 1

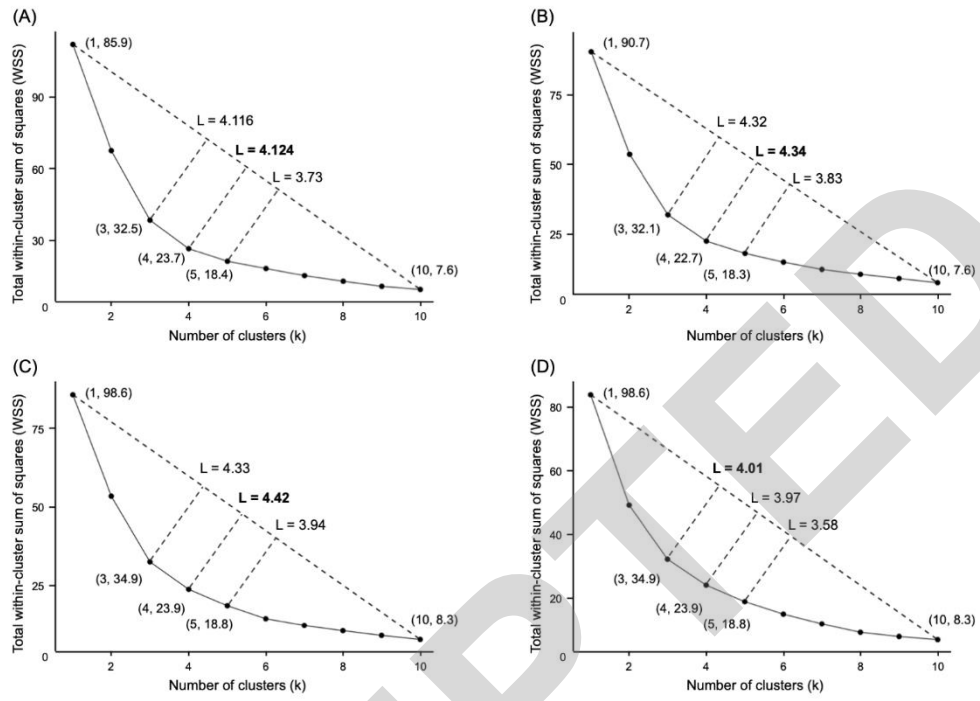


Figure 2

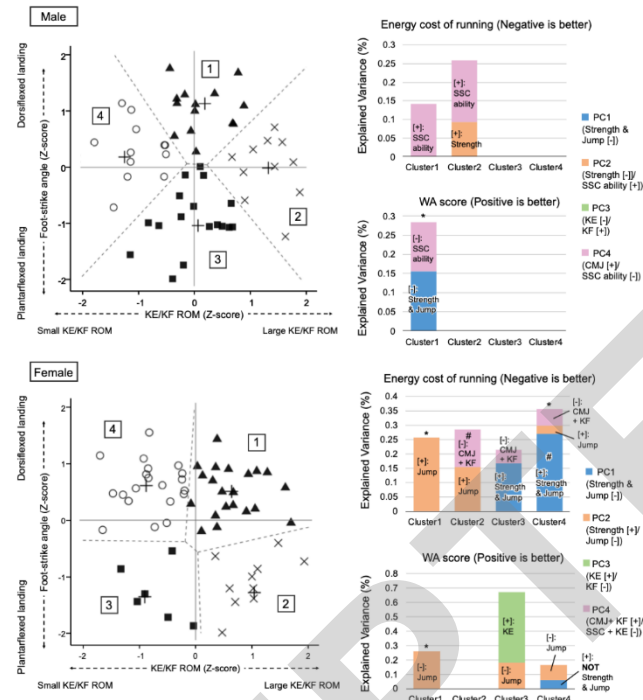


Figure 3

